Learned Latent Curves and the Hyperspherical-Harmonic Variant

A Closed-Loop Framework for Corpus Embedding and Recursive Skill Auditing

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Abstract. We describe a closed-loop framework for auditing and prioritizing improvement work across a corpus of engineering artifacts (software skills, self-documentation files), and we document a sphere-aware extension of its curve-fitting stage. The framework has three composed techniques: (1) a learned latent curve, which re-expands a single 1-D ordering coordinate t into a fixed-width D-dimensional embedding via a Fourier basis with learned frequencies; (2) curve-guided recursive self-improvement, which uses sparse cells of the fitted curve as a prioritization lens for gap-mapping and bounded fixpoint-style editing cycles; and (3) a self-documentation variant of the same pipeline, retargeted at heterogeneous memory-file corpora. We give the governing equations for each stage, the empirical validation obtained on a corpus of software skills (N = 49 to 70), and we introduce a fourth technique — the hyperspherical-harmonic curve — which replaces the flat $[0,1]^2$ parameter manifold of (1) with the Riemann sphere S^2 (the 2-sphere, with a generalization sketch to S^N). The variant uses the canonical orthonormal hyperspherical-harmonic basis and adds a learned Möbius ϕ_θ PSL(2,ℂ) reparameterization of the domain. We evaluate the variant against a capacity-matched flat Fourier baseline on two splits of the corpus, report absolute holdout R^2 and the matched-parameter delta, and identify five open items that bound the claim.

1 Introduction

Problem statement.

Given N items in a K-dimensional feature space, find (i) a scalar ordering coordinate t_i [0,1] for each item and (ii) a fixed-width D-dimensional embedding z_i $\mathbb{R}^D$ such that interpolation between adjacent items in the ordering is well defined and sparse regions of the ordering are detectable. This arises in corpus-audit tasks: a practitioner needs to know which parts of an ordering are thin and need more work.

This paper.

The paper has two halves. Sections 2–4 give the family background — the flat learned-latent-curve, the curve-guided recursive self-improvement pipeline that consumes it, and the self-documentation offshoot — at the level of governing equations and empirical context. Section 5 introduces the new contribution: a hyperspherical-harmonic curve on the Riemann sphere S^2 with a learned Möbius reparameterization, the sphere-aware analogue of the flat curve. Sections 6–9 give the dataset, evaluation protocol, results, and discussion for the variant. We do not claim the variant is a strict improvement over the flat curve in absolute terms; we report the absolute holdout R^2 alongside the matched-parameter delta and identify where the result holds and where it does not.

2 Background: The Learned-Latent-Curve Family

2.1 Flat curve model

For output dimension j = 1, …, D (canonically D = 384) and 1-D coordinate t [0,1], the model is

$$z_j(t) = a_{j0} + \Sigma_{m=1}^{k} (a_{j,m} \sin(2\pi f_m t) + b_{j,m} \cos(2\pi f_m t)) \quad (1)$$

with k shared learned frequencies $f_1, \ldots, f_k$ and per-output coefficients $a_{j,m}$, $b_{j,m}$. Stacked over j, this is a curve $\gamma : \mathbb{R}$ $\mathbb{R}^D$. Writing the design vector

$$\phi(t) = [1, \sin(2\pi f_1 t), \cos(2\pi f_1 t), \ldots, \sin(2\pi f_k t), \cos(2\pi f_k t)] \;\; \mathbb{R}^{1+2k} \quad (2)$$

the model is $\gamma(t) = C\,\phi(t)$ with coefficient matrix C $\mathbb{R}^{D \times (1+2k)}$, giving a parameter count of

$$P = k + D(1+2k) \quad (3)$$

At D = 384, k = 8: P = 8 + 384 × 17 = 6,536 parameters.

2.2 Closed-form coefficient solve

With the frequencies f held fixed, the linear coefficient solve is the Tikhonov ridge

$$C = (\Phi\Phi + \lambda I)^{-1}\, \Phi\, Z \quad (4)$$

where Φ $\mathbb{R}^{N \times (1+2k)}$ stacks the design vector and Z $\mathbb{R}^{N \times D}$ stacks the target vectors. This is the sanity floor for any gradient-descent fit at the same frequencies: a fit worse than this at the same f means the optimizer failed, not the model.

2.3 Frequency parameterization

Frequencies are stored in an unconstrained form and mapped through a softplus to guarantee strict positivity and finite gradients near zero:

$$f_m = \mathrm{softplus}(\tilde{f}_m) = \log(1 + \exp(\tilde{f}_m)), \quad f_m > 0 \quad (5)$$

2.4 Losses

Let t $\mathbb{R}^N$, Z $\mathbb{R}^{N \times D}$ the targets, and $\hat{Z} = \gamma(t)$ the fitted curve evaluated at the training coordinates. The total objective is

$$= {}_{rec} + {}_{freq} + {}_{smooth} \quad (6)$$

The reconstruction term ${}_{rec} = \mathrm{mean}((\hat{Z} - Z)^2)$ measures fit. The frequency-magnitude regulariser ${}_{freq} = \lambda_f \cdot \mathrm{mean}(f^2)$ with λ_f 10^{-4} keeps the learned frequencies from collapsing to the Nyquist limit. The smoothness regulariser

$${}_{smooth} = \lambda_s \cdot \mathrm{mean}((\gamma(t_{grid})[2:] - 2\gamma(t_{grid})[1:-1] + \gamma(t_{grid})[:-2])^2) \quad (7)$$

with t_{grid} a dense grid of ~512 points spanning the coordinate range and λ_s 10^{-3}, penalises the second-difference of the curve between observed points, where reconstruction loss alone is blind.

2.5 Obtaining the coordinate t

The default pipeline sets t from the top principal component of a z-scored N × D_{feat} feature matrix, mapped to [0,1]; a rank-uniformized variant replaces raw PC1 scores by their ranks to improve design-matrix conditioning. The explained-variance ratio of PC1 (or PC1+PC2 for a 2-D surface variant) is recorded as an explicit go/no-go gate, with PC1 0.40 treated as evidence of usable low-rank structure.

3 Curve-Guided Recursive Self-Improvement

3.1 Pipeline

The pipeline has four stages plus a re-fit verification step:

1. 1. Fit the curve (Stage 1) — reuse Section 2.
2. 2. Detect sparse cells (Stage 2) — a uniform 0.05 × 0.05 grid in $[0,1]^2$ partitions the parameter space; cells with 1 item are isolated.
3. 3. Dispatch gap-mapping (Stage 3) — a fresh-context gap-mapping subagent (focused on a single sparse-cell item, never the whole corpus) is dispatched per isolated item.
4. 4. Apply recursive-self-improvement cycles (Stage 4) — capped at 3 cycles per run, with the fixpoint rule no new substantive gaps AND old gaps closed AND no new anti-patterns.

5. 5. Re-fit and verify (Stage 5) — Stage 1 is re-run after all cycles; the closed-loop claim is verified by comparing the sparse-cell count before and after.

The success metric is

$$\Delta_{sparse} = |sparse\_cells|_{post} - |sparse\_cells|_{pre} < 0 \quad (8)$$

3.2 Concrete cell definition

For a 2-D surface fit, coordinates are reduced to (u, v) $[0,1]^2$ via PC1+PC2 of a binary primitive-coverage matrix C $\{0,1\}^{N \times 9}$ lifted to D = 384 by a fixed seeded orthonormal projection Q:

$$Z = C\,Q, \quad (u_i, v_i) = PCA_{1,2}(Z)_i \quad (9)$$

The $[0,1]^2$ plane is discretized into a 0.05 × 0.05 grid (21 × 21 = 441 cells). For a cell c = (u, v),

$$neighbors(c) := \{ i \;\; corpus : \|(u_i, v_i) - c\| \;\; r \}, \quad (10)$$

$$is\_sparse(c) := |neighbors(c)| = 0 \quad (11)$$

with default radius r = 0.05.

4 Self-Documentation Variant

The self-documentation offshoot retargets Eqs. 9–11 at heterogeneous memory-file corpora (`SELF.md`, `SELF-CHANGELOG.md`, and additional preference / rule files), under the granularity rule that each version is one corpus item (a changelog entry, a memory-file section, or a named row). Each corpus receives its own 9-D binary primitive basis rather than sharing one basis across structurally distinct document types, since a single shared basis was found to flatten the per-corpus signal. Near-constant columns (coverage > 0.90 or < 0.10) are dropped before the PCA lift in Eq. 9.

5 Notation

D — embedding dimension (default 384).
N — corpus size; N_{train}, $N_{holdout}$ — split sizes.
k — number of Fourier frequencies in the flat baseline (default 8).
L — maximum harmonic degree in the spherical basis (default 3).
x S^2 $\mathbb{R}^3$ — input point on the Riemann sphere (default; S^N generalization sketched in Section 6.1).
$Y^{S^2}_{\ell,m}(x)$ — real spherical harmonic of degree ℓ, order m, on S^2.
$\phi_\theta(z) = (az+b)/(cz+d)$ — Möbius transformation with ad-bc = +1, parameterised by θ.
$\chi(z_1,z_2;z_3,z_4) = (z_1-z_3)(z_2-z_4)/((z_1-z_4)(z_2-z_3))$ — cross-ratio.

6 The Hyperspherical-Harmonic Curve

6.1 Model

For x S^2 $\mathbb{R}^3$ (the Riemann sphere, with a generalization sketch to S^N given at the end of this section),

$$\gamma(x) = b + \Sigma_{\ell=0}^{L} \Sigma_m a_{\ell,m} Y^{S^2}_{\ell,m}(\phi_\theta(x)) \quad (12)$$

where b $\mathbb{R}^D$ is the per-dim bias, $a_{\ell,m}$ $\mathbb{R}^D$ are the per-dim per-basis coefficients, and ϕ_θ is the Möbius reparameterization.

The parameter count is D · n_{basis} + D + $n_{Möbius}$, where $n_{basis} = \Sigma_{\ell=0}^{L}(2\ell+1)$ and $n_{Möbius}$ is the real dimension of PSL(2,ℂ) (Section 6.3). At L = 3 on S^2, n_{basis} = 16; at D = 384: P = 384·16 + 384 + 6 = 6,534.

Generalization to S^N: for N 3 the same construction applies with the canonical S^N hyperspherical-harmonic basis $\{Y^{S^N}_{\ell,m}\}$ — eigenfunctions of the Laplace–Beltrami operator on S^N with eigenvalues $-\ell(\ell+N-1)$. The Möbius reparameterization is specific to S^2 (it is the automorphism group of Ĉ $\mathbb{C}P^1$ S^2); for higher N the corresponding parameterization would be the isometry group of the sphere.

6.2 Real spherical harmonics on S^2

The real spherical harmonics $Y^{S^2}_{\ell,m}$ on S^2 are constructed via explicit Legendre polynomials and the cos/sin split:

$$Y^c_{\ell,0}(\theta, \phi) = N_\ell^{\,0} \cdot P_\ell^{\,0}(\cos\theta)$$

$$Y^c_{\ell,m}(\theta, \phi),\ \ m > 0 = 2 \cdot N_\ell^{\,m} \cdot P_\ell^{\,m}(\cos\theta) \cdot \cos(m\phi)$$

$$Y^c_{\ell,-m}(\theta, \phi),\ \ m > 0 = 2 \cdot N_\ell^{\,m} \cdot P_\ell^{\,m}(\cos\theta) \cdot \sin(m\phi)$$

with $N_\ell^{\,m} = [(2\ell+1)/(4\pi) \cdot (\ell-m)!/(\ell+m)!]$ the standard normalisation. At $L = 3$ on S^2 the basis has $1 + 3 + 5 + 7 = 16$ functions; at $L = 3$ on S^3 the basis has 30 functions.

6.3 Möbius reparameterization

$\phi_\theta(z) = (az+b)/(cz+d)$ with a,b,c,d $\mathbb{C}$ and $ad-bc = +1$. The four complex coefficients a,b,c,d carry 8 real components. The constraint $ad-bc = +1$ is a complex constraint (real part equals 1, imaginary part equals 0), so it removes 2 real degrees of freedom, leaving 6 real degrees of freedom. The $PSL(2,\mathbb{C})$ identification (matrices M and -M map to the same Möbius transformation) is a discrete identification and does not further reduce the real dimension. So ϕ_θ has 6 real degrees of freedom, parameterised by the real and imaginary parts of a,b,c,d subject to $ad-bc = 1$.

We refine θ via L-BFGS-B with the closed-form ridge (Eq. 4) re-solved at each step. The basis is fixed; only the domain is reparameterized. Because every ϕ $PSL(2,\mathbb{C})$ preserves the cross-ratio identically by definition, the cross-ratio check verifies implementation consistency: it fails if ϕ_θ is miscoded, not if the learned map is a poor fit.

7 Dataset and Evaluation Protocol

7.1 Dataset

The corpus consists of software-skill specifications (engineering artifacts). Each item has a 9-dimensional binary feature vector recording which of nine primitive capabilities the skill implements. We consider two splits:

Split A (49 items): the first 49 items alphabetically. Train $N_{train} = 35$, holdout $N_{holdout} = 14$.
Split B (70 items): all items. Train $N_{train} = 49$, holdout $N_{holdout} = 21$.

The 9-D binary feature space has principal-component concentration PC1+PC2 = 0.652 at Split A and 0.548 at Split B; both clear the PC1 0.40 gate (Section 2.5). The split sizes are chosen because (i) a curve is the honest model for small N and (ii) the $S^2/L = 3$ basis needs at least N 48 items to fit.

7.2 Baseline

The capacity-matched baseline is a flat Fourier curve on $[0,1]^2$ with $k = 2$ (the 2-D tensor-product extension of Eq. 1), giving $5 \times 5 = 25$ basis functions and 9,984 parameters. Both models are fitted with the closed-form ridge (Eq. 4) and the same ridge regularisation λ.

7.3 Evaluation metric

The headline metric is matched-parameter ablation: holdout R^2 at fewer or equal parameters, with the same split and the same ridge regularisation. We report the absolute holdout R^2 (not just the delta) so the result is interpretable against the corpus-mean baseline ($R^2 = 0$).

7.4 Reproducibility

The basis construction is deterministic given (ℓ, m). The Möbius refinement uses L-BFGS-B; the initial point is θ_0 with $a = d = 1$, $b = c = 0$ (identity Möbius). The ridge λ is fixed across both models. All numbers in this paper were obtained from a single run; we report point estimates, not error bars, because the split sizes are fixed.

8 Results

8.1 Flat-curve context (earlier fits)

Earlier flat-curve fits on the same family of corpora (Table 1) show the surface the hyperspherical variant is compared against. The headline flat fit on a 211-item skill corpus with a 2-D surface (PC1+PC2 = 0.4655) reached holdout R^2 = +0.4655; the raw-content target on a 62-item subset failed at R^2 = -0.155; the sentence-transformer target on the 211-item corpus returned R^2 between -0.005 and +0.130 (target failure). Curve-guided RSI on a 69-skill corpus reached R^2 = +0.2244 on the curve-fit metric. The self-doc corpus reaches much higher R^2 (0.7041 to 0.9917) but on structurally different per-file primitive bases.

Corpus / fit	N	PC1(+PC2)	Holdout R^2	Status
Skill corpus, raw-content target	62	0.40 (1-D)	-0.155	target fail
Skill corpus, primitive-coverage (1-D)	213	0.40 (1-D)	+0.183	pass
Skill corpus, primitive-coverage (2-D)	211	0.4655	+0.4655	headline pass
Skill corpus, sentence-transformer	211	0.0955	-0.005 to +0.130	target fail
Curve-guided RSI validation	69	0.4615	+0.2244	pass
Self-doc corpus (CHANGELOG)	17	0.9164	+0.9917	pass
Self-doc corpus (SELF.md)	51	0.6952	+0.7041	pass
Self-doc corpus, combined	154	0.6479	+0.7877	pass

Table 1. Flat-curve fit quality across corpora. PC1(+PC2) is the explained-variance gate (Section 2.5); holdout R^2 is measured against held-out items refit from their t coordinate alone. These are the surface the hyperspherical-harmonic variant in Section 6 is compared against.

8.2 Hyperspherical: Split A (49 items)

Hyperspherical model: R^2_{holdout} = +0.379 (6,534 parameters).
Flat Fourier baseline: R^2_{holdout} = -0.359 (9,984 parameters).
Delta (spherical - flat): +0.737.

The hyperspherical model outperforms the flat baseline at fewer parameters. The absolute R^2 is positive on this split.

8.3 Hyperspherical: Split B (70 items)

Hyperspherical model: R^2_{holdout} = -0.236 (6,534 parameters).
Flat Fourier baseline: R^2_{holdout} = -0.756 (9,984 parameters).
Delta (spherical - flat): +0.520.

Both models perform worse than predicting the corpus mean (R^2 = 0) on this split. The hyperspherical model is less negative than the flat baseline; the relative comparison is positive. The absolute result is negative.

8.4 Calibration checks

- Spectral mass $\rho = \Sigma_{\ell 1} \|a_{:,\ell}\|^2 / \Sigma_{\ell 0} \|a_{:,\ell}\|^2$ 0.10: measured 0.977 (A) / 0.983 (B).
- High-degree mass $\Sigma_{\ell > L/2} \|a_{:,\ell}\|^2$ / total 0.40: measured 0.206 (A) / 0.178 (B).
- Cross-ratio check on 100 held-out 4-tuples: max residual 3.08×10^{-7}. Consistent with float64 noise; every $PSL(2,\mathbb{C})$ element preserves χ exactly, so the residual measures implementation precision, not fit quality.
- Möbius refinement (train R^2): identity 0.9125 refined 0.9211, Δ = +0.009. Train-only; no holdout effect measured.

9 Discussion

9.1 What this result does and does not show

The relative comparison (spherical vs flat on the same split) is positive on both splits. The absolute R^2 is positive only on Split A; on Split B, both models are worse than the corpus mean. The result supports the claim that, on this corpus, a spherical parameter manifold is a better inductive bias than a flat one; it does not support the claim that either model is a good fit in absolute terms.

9.2 Why a sphere?

The flat $[0,1]^2$ manifold has zero scalar curvature, trivial topology, and trivial holonomy. The Riemann sphere S^2 has constant positive scalar curvature +2, simply-connected topology (χ = 2), and non-trivial holonomy. For corpora whose intrinsic geometry is well-modelled by S^2, the curved parameter manifold is a better inductive bias. The deltas are properties of the chosen domain, not of the fit, and are motivation rather than evidence; the ablation is the evidence.

9.3 Why Möbius?

PSL(2,$\mathbb{C}$) is the full automorphism group of the Riemann sphere $\hat{\mathbb{C}}$ $\mathbb{C}P^1$ S^2 and the 6-parameter family of invertible conformal domain warps on it. The Möbius refinement is not isometric: only the subgroup PSU(2) SO(3) preserves the round metric. This non-isometry is what makes the refinement able to change the fit at all (a purely isometric reparameterization could not).

9.4 Limitations

(1) The implementation uses explicit Legendre + cos/sin split. Production could swap to a maintained basis library. (2) We evaluate on a single target (binary primitive coverage). The result may not transfer to other targets (raw content, sentence-transformer embeddings); Table 1 shows the raw-content and sentence-transformer targets failing even on the flat curve. (3) Split B's negative absolute R^2 indicates the corpus is at the edge of the curve model's honest range. We do not know where that edge is. (4) S^3/L = 3 (30 basis functions) would need N 90 items AND PC3 0.08 to use. The current corpus is below both thresholds. (5) The matched-parameter ablation is a relative result; both arms fail the absolute holdout-R^2 > 0 gate at Split B. The headline is the delta, not the absolute.

10 Related Work

We did not identify a prior work that combines hyperspherical-harmonic basis, learned Möbius reparameterization, and corpus-fit evaluation on a software-skill corpus. We searched for related work using 11 keyword patterns across the closest candidates:

arXiv:2601.20528 (Durastanti, 2026) — "Spectral Bayesian Regression on the Sphere." Covers Fourier-on-S^2 as Bayesian regression with statistical-theory focus; posterior contraction rates, Laplace–Beltrami smoothing splines. Does not cover: corpus fit, learned Möbius reparameterization, sparse-cell detection.

OpenReview `g6UqpVislvH` (ICLR 2022 submission) — "Generalized Fourier Features for Coordinate-Based Learning on Manifolds." Covers positional encoding for NeRF / panorama / SO(3) using spherical harmonics + SO(2)/SO(3) rotation shifts. Does not cover: corpus fit, PSL(2,$\mathbb{C}$) Möbius, sparse-cell detection.

The application-layer composition (hyperspherical basis + Möbius + corpus fit) appears to be a gap in the published literature as of this writing.

10.1 Unrelated prior work

Eq. 1 shares no derivation with the weight-space reparameterization proposed in arXiv:2607.09967 (Smith, 2026, "Learning in Curved Weight Space: Exponential-Linear Weight Reparameterization for Improved Optimization"), which maps a single raw neural-network weight scalar w through a sign-aware symmetric-exponential/linear pathway pair to obtain an effective weight w_{eff} for optimizer training, validated by training-step reduction on autoregressive language models. The present work instead maps a per-item ordering coordinate t through a Fourier basis to obtain a fixed-width corpus embedding, validated by holdout reconstruction accuracy on document corpora. The overlap is limited to the word “curve”/“curvature” and the presence of learnable shape parameters; neither the domain, the basis function family, nor the fitting objective is shared. The two are unrelated by construction, not merely by citation omission.

Within the embedding literature, Eq. 1 is a fitted generalization of two established fixed-basis constructions: random Fourier features use fixed random frequencies as a kernel approximation, and transformer positional encodings use fixed geometric frequencies with no learned coefficients. The present model learns both the frequencies and the coefficients from the target embeddings directly.

11 Conclusion

The learned-latent-curve family gives a closed-form, auditable, interpolatable representation for small-N corpora with low-rank ordering structure; composed with sparse-cell detection and a bounded gap-closing loop, it converts a one-shot corpus audit into a measurable, repeatable improvement process. The hyperspherical-harmonic curve is a candidate Stage-1 method for this family: on the splits we tested, it outperforms the capacity-matched flat Fourier baseline at fewer parameters, with the absolute holdout R^2 positive on the smaller split ($N = 49$) and negative on the larger ($N = 70$). The method's honest range is bounded by the corpus size; we describe the method, the dataset, and the protocol in full, and we identify five open items that bound the claim.

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